

分布式计算实验 2

1. 实验要求

按照实验指南说明在树莓派集群上进行 Hadoop 词频统计实验，可自行准备不同规模的输入数据，记录不同规模输入数据对应的 map-reduce 过程执行时间，分析随工作负载的变化相应执行时间的变化趋势。

2. 实验步骤

测试文件 poem1

```
renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -copyFromLocal poem01 /user/renjianxin/wordcount/input
renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -ls /user/renjianxin/wordcount/input
Found 1 items
-rw-r--r-- 3 renjianxin hadoop 725 2026-04-30 13:36 /user/renjianxin/wordcount/input/poem01
```

```
renjianxin@raspi00:~/hadoop-examples $ time hadoop jar wc.jar WordCount /user/renjianxin/wordcount/input /user/renjianxin/wordcount/output
2026-04-30 13:42:51,371 INFO client.RMPProxy: Connecting to ResourceManager at raspi00/192.168.86.10:8032
2026-04-30 13:42:53,672 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-30 13:42:53,704 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/renjianxin/.staging/job_1777467880002_0002
2026-04-30 13:42:54,766 INFO input.FileInputFormat: Total input files to process : 1
2026-04-30 13:42:55,341 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-30 13:42:55,614 INFO Configuration.deprecation: yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-04-30 13:42:56,146 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777467880002_0002
2026-04-30 13:42:56,151 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-30 13:42:57,152 INFO conf.Configuration: resource-types.xml not found
2026-04-30 13:42:57,153 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-30 13:42:57,190 WARN mapred.YARNRunner: Configuration yarn.app.mapreduce.am.resource.memory-mb=512Mi is overriding the yarn.app.mapreduce.am.resource.mb=1536 configuration
2026-04-30 13:42:57,450 INFO impl.YarnClientImpl: Submitted application application_1777467880002_0002
2026-04-30 13:42:57,663 INFO mapreduce.Job: The url to track the job: http://raspi00:8088/proxy/application_1777467880002_0002/
2026-04-30 13:42:57,666 INFO mapreduce.Job: Running job: job_1777467880002_0002
2026-04-30 13:43:23,740 INFO mapreduce.Job: Job job_1777467880002_0002 running in uber mode : false
2026-04-30 13:43:23,748 INFO mapreduce.Job: map 0% reduce 0%
2026-04-30 13:43:36,634 INFO mapreduce.Job: map 100% reduce 0%
2026-04-30 13:43:50,927 INFO mapreduce.Job: map 100% reduce 100%
2026-04-30 13:43:51,982 INFO mapreduce.Job: Job job_1777467880002_0002 completed successfully
2026-04-30 13:43:52,587 INFO mapreduce.Job: Counters: 53
File System Counters
  FILE: Number of bytes read=1115
  FILE: Number of bytes written=428067
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
```

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```

2026-04-30 13:43:52,587 INFO mapreduce.Job: Counters: 53
  File System Counters
    FILE: Number of bytes read=1115
    FILE: Number of bytes written=428067
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=843
    HDFS: Number of bytes written=702
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
  Job Counters
    Launched map tasks=1
    Launched reduce tasks=1
    Rack-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=10258
    Total time spent by all reduces in occupied slots (ms)=21942
    Total time spent by all map tasks (ms)=10258
    Total time spent by all reduce tasks (ms)=10971
    Total vcore-milliseconds taken by all map tasks=10258
    Total vcore-milliseconds taken by all reduce tasks=10971
    Total megabyte-milliseconds taken by all map tasks=2626048
    Total megabyte-milliseconds taken by all reduce tasks=5617152
  Map-Reduce Framework
    Map input records=28
    Map output records=151
    Map output bytes=1274
    Map output materialized bytes=1115
    Input split bytes=118
    Combine input records=151
    Combine output records=102
    Reduce input groups=102
    Reduce shuffle bytes=1115
    Reduce input records=102
    Reduce output records=102
    Spilled Records=204
    Shuffled Maps =1
    Failed Shuffles=0
    Merged Map outputs=1
    GC time elapsed (ms)=764
    CPU time spent (ms)=3010
    Physical memory (bytes) snapshot=252600320
    Virtual memory (bytes) snapshot=950091776
    Total committed heap usage (bytes)=137891840
    Peak Map Physical memory (bytes)=176447488
    Peak Map Virtual memory (bytes)=366108672
    Peak Reduce Physical memory (bytes)=76152832
    Peak Reduce Virtual memory (bytes)=583983104
  Shuffle Errors

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  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=725
  File Output Format Counters
    Bytes Written=702

```

```

real    1m8.255s
user    0m13.355s
sys     0m0.758s

```

```

renjianxin@raspi00:~/hadoop-examples $ hadoop fs -rm -r /user/renjianxin/wordcount/output
Deleted /user/renjianxin/wordcount/output
renjianxin@raspi00:~/hadoop-examples $ hadoop fs -rm -r /user/renjianxin/wordcount/input
Deleted /user/renjianxin/wordcount/input

```

测试 medium.txt

```
renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -mkdir -p /user/renjianxin/wordcount/input
renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -copyFromLocal medium.txt /user/renjianxin/wordcount/input
renjianxin@raspi00:~/hadoop-examples/testdata $ cd ..
renjianxin@raspi00:~/hadoop-examples $ time hadoop jar wc.jar WordCount /user/renjianxin/wordcount/input /user/renjianxin/wordcount/output
2026-04-30 14:04:11,112 INFO client.RMPProxy: Connecting to ResourceManager at raspi00/192.168.86.10:8032
2026-04-30 14:04:12,996 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-30 14:04:13,045 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/renjianxin/.staging/job_1777467880002_0003
2026-04-30 14:04:14,029 INFO input.FileInputFormat: Total input files to process : 1
2026-04-30 14:04:14,531 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-30 14:04:14,655 INFO Configuration.deprecation: yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-04-30 14:04:15,122 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777467880002_0003
2026-04-30 14:04:15,127 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-30 14:04:16,039 INFO conf.Configuration: resource-types.xml not found
2026-04-30 14:04:16,041 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-30 14:04:16,074 WARN mapred.YARNRunner: Configuration yarn.app.mapreduce.am.resource.memory-mb=512Mi is overriding the yarn.app.mapreduce.am.resource.mb=1536 configuration
2026-04-30 14:04:16,437 INFO impl.YarnClientImpl: Submitted application application_1777467880002_0003
2026-04-30 14:04:16,599 INFO mapreduce.Job: The url to track the job: http://raspi00:8088/proxy/application_1777467880002_0003/
2026-04-30 14:04:16,601 INFO mapreduce.Job: Running job: job_1777467880002_0003
2026-04-30 14:04:36,443 INFO mapreduce.Job: Job job_1777467880002_0003 running in uber mode : false
2026-04-30 14:04:36,449 INFO mapreduce.Job: map 0% reduce 0%
2026-04-30 14:04:52,910 INFO mapreduce.Job: map 100% reduce 0%
2026-04-30 14:05:13,308 INFO mapreduce.Job: map 100% reduce 100%
2026-04-30 14:05:14,359 INFO mapreduce.Job: Job job_1777467880002_0003 completed successfully
2026-04-30 14:05:14,895 INFO mapreduce.Job: Counters: 53
File System Counters
  FILE: Number of bytes read=196358
  FILE: Number of bytes written=818553
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1000122
  HDFS: Number of bytes written=143369
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
Job Counters
```

```
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=14142
  Total time spent by all reduces in occupied slots (ms)=31408
  Total time spent by all map tasks (ms)=14142
  Total time spent by all reduce tasks (ms)=15704
  Total vcore-milliseconds taken by all map tasks=14142
  Total vcore-milliseconds taken by all reduce tasks=15704
  Total megabyte-milliseconds taken by all map tasks=3620352
  Total megabyte-milliseconds taken by all reduce tasks=8040448
Map-Reduce Framework
  Map input records=13018
  Map output records=175914
  Map output bytes=1656670
  Map output materialized bytes=196358
  Input split bytes=122
  Combine input records=175914
  Combine output records=13807
  Reduce input groups=13807
  Reduce shuffle bytes=196358
  Reduce input records=13807
  Reduce output records=13807
  Spilled Records=27614
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=794
  CPU time spent (ms)=7670
  Physical memory (bytes) snapshot=253218816
  Virtual memory (bytes) snapshot=950243328
  Total committed heap usage (bytes)=137891840
  Peak Map Physical memory (bytes)=176762880
  Peak Map Virtual memory (bytes)=366108672
  Peak Reduce Physical memory (bytes)=76455936
  Peak Reduce Virtual memory (bytes)=584134656
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=1000000
File Output Format Counters
  Bytes Written=143369
```

```

Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
File Input Format Counters
    Bytes Read=1000000
File Output Format Counters
    Bytes Written=143369

real    1m10.542s
user    0m13.065s
sys     0m0.676s

```

```

renjianxin@raspi00:~/hadoop-examples $ hadoop fs -rm -r /user/renjianxin/wordcount/input/*
Deleted /user/renjianxin/wordcount/input/medium.txt
renjianxin@raspi00:~/hadoop-examples $ hadoop fs -rm -r /user/renjianxin/wordcount/out
^Crenjianxin@raspi00:~/hadoop-examples $ hadoop fs -rm -r /user/renjianxin/wordcount/output
Deleted /user/renjianxin/wordcount/output

```

测试 big.txt

```

renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -copyFromLocal big.txt /user/renjianxin/wordcount/input
renjianxin@raspi00:~/hadoop-examples/testdata $ hadoop fs -ls /user/renjianxin/wordcount/input
Found 1 items
-rw-r--r-- 3 renjianxin hadoop 6488666 2026-04-30 14:17 /user/renjianxin/wordcount/input/big.txt
renjianxin@raspi00:~/hadoop-examples/testdata $ cd ..
renjianxin@raspi00:~/hadoop-examples $ time hadoop jar wc.jar WordCount /user/renjianxin/wordcount/input /user/renjianxin/wordcount/output
2026-04-30 14:18:59,087 INFO client.RMProxy: Connecting to ResourceManager at raspi00/192.168.86.10:8032
2026-04-30 14:19:00,971 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-30 14:19:01,024 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/renjianxin/.staging/job_1777467880002_0004
2026-04-30 14:19:02,079 INFO input.FileInputFormat: Total input files to process : 1
2026-04-30 14:19:02,599 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-30 14:19:02,730 INFO Configuration.deprecation: yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-04-30 14:19:03,189 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777467880002_0004
2026-04-30 14:19:03,194 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-30 14:19:04,103 INFO conf.Configuration: resource-types.xml not found
2026-04-30 14:19:04,104 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-30 14:19:04,142 WARN mapred.YARNRunner: Configuration yarn.app.mapreduce.am.resource.memory-mb=512Mi is overriding the yarn.app.mapreduce.am.resource.mb=1536 configuration
2026-04-30 14:19:04,430 INFO impl.YarnClientImpl: Submitted application application_1777467880002_0004
2026-04-30 14:19:04,595 INFO mapreduce.Job: The url to track the job: http://raspi00:8088/proxy/application_1777467880002_0004/
2026-04-30 14:19:04,598 INFO mapreduce.Job: Running job: job_1777467880002_0004
2026-04-30 14:19:24,439 INFO mapreduce.Job: Job job_1777467880002_0004 running in uber mode : false
2026-04-30 14:19:24,444 INFO mapreduce.Job: map 0% reduce 0%
2026-04-30 14:19:55,410 INFO mapreduce.Job: map 67% reduce 0%
2026-04-30 14:20:10,765 INFO mapreduce.Job: map 100% reduce 0%
2026-04-30 14:20:31,133 INFO mapreduce.Job: map 100% reduce 100%
2026-04-30 14:20:32,196 INFO mapreduce.Job: Job job_1777467880002_0004 completed successfully
2026-04-30 14:20:32,757 INFO mapreduce.Job: Counters: 53

```



```

File System Counters
  FILE: Number of bytes read=523929
  FILE: Number of bytes written=1473695
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=6488785
  HDFS: Number of bytes written=390040
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Rack-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=40736
  Total time spent by all reduces in occupied slots (ms)=32444
  Total time spent by all map tasks (ms)=40736
  Total time spent by all reduce tasks (ms)=16222
  Total vcore-milliseconds taken by all map tasks=40736
  Total vcore-milliseconds taken by all reduce tasks=16222
  Total megabyte-milliseconds taken by all map tasks=10428416
  Total megabyte-milliseconds taken by all reduce tasks=8305664
Map-Reduce Framework
  Map input records=128457
  Map output records=1106820
  Map output bytes=10616994
  Map output materialized bytes=523929
  Input split bytes=119
  Combine input records=1106820
  Combine output records=35938
  Reduce input groups=35938
  Reduce shuffle bytes=523929
  Reduce input records=35938
  Reduce output records=35938
  Spilled Records=71876
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=1108
  CPU time spent (ms)=30180
  Physical memory (bytes) snapshot=253423616
  Virtual memory (bytes) snapshot=951287808
  Total committed heap usage (bytes)=137891840
  Peak Map Physical memory (bytes)=177000448
  Peak Map Virtual memory (bytes)=367292416
  Peak Reduce Physical memory (bytes)=76423168
  Peak Reduce Virtual memory (bytes)=583995392
Shuffle Errors
  BAD_ID=0

```

```

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=6488666
File Output Format Counters
  Bytes Written=390040

```

```

real    1m40.415s
user    0m13.481s
sys     0m0.813s

```

```

renjianxin@raspi00:~/hadoop-examples $

```

3. 结果分析

实验组	输入文件	数据规模	总执行时间
1	poem01	725B	1m 8.255s
2	medium.txt	1000000B	1m 10.542s
3	big.txt	6488666B	1m 40.415s

在处理极小规模数据时，作业执行时间主要由 Hadoop 框架的启动开销决定。即使数据量成百上千倍增长，只要未达到计算密集型阈值，总执行时间保持稳定。当数据规模进一步扩大到 6.5MB 时，执行时间的增长幅度显著超过了数据量的增长幅度。这表明树莓派集群的硬件资源开始成为性能瓶颈。Shuffle 阶段的数据传输和磁盘读写延迟导致了超线性的性能下降。

实验结果表明，基于 ARM 架构和 SD 卡存储的树莓派集群适合处理中小规模数据或用于算法逻辑验证。对于大规模数据处理，硬件 I/O 瓶颈会导致性能急剧下降。